



CASE STUDY

HYDROPNEUMATIC



INDUSTRY - WATER

OBJECTIVE

Implement optimized solutions with Best-In-Class products, software development, and project management methodologies to maximize profits from machines, processes, and plants





CHALLENGES

- 1 Cost-efficient booster pump controller
- 2 Minimizing downtime during maintenance and troubleshooting
- 3 Maintaining equal run-hours for all pumps
- 4 Connectivity with the existing Building Management System (BMS) network
- 5 Meeting regulatory and safety requirements



- 6 Compatibility with varying pump capacities
- 7 Adaptability to different operating conditions
- 8 Pressure control with PID
- 9 Ensuring reliable and accurate sensor readings
- 10 Efficient energy utilization

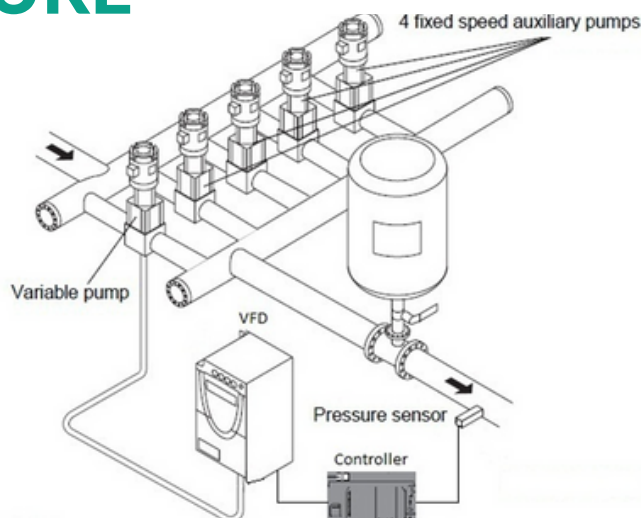




SOLUTION

- 1 Utilizing cost-effective components without compromising performance
- 2 Implementing advanced PID control algorithms for precise pressure regulation
- 3 Incorporating intelligent algorithms to balance pump run-hours
- 4 Integrating with BMS using industry-standard protocols for seamless connectivity
- 5 Implementing energy-saving features and optimization strategies
- 6 Offering a range of controller options to accommodate different pump capacities
- 7 Providing customizable settings and set points to meet diverse operational needs
- 8 Building robust fault diagnosis and remote troubleshooting capabilities
- 9 Conducting rigorous sensor calibration and health checks for accurate readings
- 10 Ensuring compliance with relevant industry standards and safety regulations

ARCHITECTURE





BENEFITS

- ✓ Reliable and cost-efficient booster pump control system
- ✓ Precise pressure control with built-in PID
- ✓ Streamlined Load Analysis through One-Click Report Generation
- ✓ Equal run-hours for all pumps, optimizing maintenance schedules
- ✓ Seamless integration with the existing BMS network
- ✓ Enhanced security with password protection
- ✓ User-friendly parameter settings
- ✓ Versatile set points for different operating conditions



📍 2/1 "Chitra-Ami",
Opp. Old RBI, Ashram Road,
Ahmedabad-380009,
Gujarat - INDIA

☎ +91 79-40036958

✉ info@adaptive-engg.com